Worksheet 5.2 — Differentiation Rules

Mathematics: Analysis & Approaches SL (IB) · Unit 5: Differential Calculus

The Big Ideas

Product, quotient, and chain rules combine to differentiate any layered expression.

- Product: $(uv)' = u'v + uv'$. Quotient: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$. Chain: $[f(g(x))]' = f'(g(x))g'(x)$.

Worked Examples

Example 1 — Product rule

Differentiate $f(x) = (3x+1)(x^2-2)$.

Solution. $u = 3x+1, v = x^2-2$: $f'(x) = 3(x^2-2) + (3x+1)(2x) = 3x^2 - 6 + 6x^2 + 2x$.

Conclusion: $f'(x) = 9x^2 + 2x - 6$.

Example 2 — Quotient rule

Differentiate $f(x) = \dfrac{2x}{x^2+4}$.

Solution. $\dfrac{2(x^2+4) - 2x(2x)}{(x^2+4)^2} = \dfrac{2x^2 + 8 - 4x^2}{(x^2+4)^2}$.

Conclusion: $f'(x) = \dfrac{8 - 2x^2}{(x^2+4)^2}$.

Example 3 — Chain rule

Differentiate $f(x) = (5x-2)^7$.

Solution. $7(5x-2)^6 \cdot 5$.

Conclusion: $f'(x) = 35(5x-2)^6$.

Example 4 — Product combined with chain

Differentiate $f(x) = x^3(2x+1)^4$, factoring your answer.

Solution. $3x^2(2x+1)^4 + x^3 \cdot 4(2x+1)^3 \cdot 2$; factor $x^2(2x+1)^3$: $3(2x+1) + 8x$.

Conclusion: $f'(x) = x^2(2x+1)^3(14x+3)$.

Example 5 — Chain rule applied twice (nested composition)

Differentiate $f(x) = \sqrt{(3x-1)^2 + 4}$ (write as a power first).

Solution. $f(x) = \left[(3x-1)^2 + 4\right]^{1/2}$; outer derivative $\frac{1}{2}[\cdot]^{-1/2}$, inner derivative $2(3x-1)(3) = 6(3x-1)$.

Conclusion: $f'(x) = \dfrac{6(3x-1)}{2\sqrt{(3x-1)^2+4}} = \dfrac{3(3x-1)}{\sqrt{(3x-1)^2+4}}$.

Example 6 — Quotient rule combined with chain

Differentiate $f(x) = \dfrac{(x+1)^2}{x-3}$.

Solution. $u = (x+1)^2, v = x-3$: $u' = 2(x+1)$. $f'(x) = \dfrac{2(x+1)(x-3) - (x+1)^2}{(x-3)^2}$.

Factor $(x+1)$ from the numerator: $(x+1)\big[2(x-3) - (x+1)\big] = (x+1)(x-7)$.

Conclusion: $f'(x) = \dfrac{(x+1)(x-7)}{(x-3)^2}$.

Example 7 — Finding where the tangent is horizontal

Find the x-value(s) where the tangent to $f(x) = x^3 - 12x + 5$ is horizontal.

Solution. Horizontal tangent means $f'(x) = 0$: $3x^2 - 12 = 0 \Rightarrow x^2 = 4$.

Conclusion: $x = 2$ or $x = -2$.

Example 8 — Product rule with three factors (via grouping)

Differentiate $f(x) = x(x+1)(x-2)$ by first expanding two factors.

Solution. Expand $(x+1)(x-2) = x^2 - x - 2$, so $f(x) = x^3 - x^2 - 2x$.

Conclusion: $f'(x) = 3x^2 - 2x - 2$.

Example 9 — Application — marginal cost

A company's cost function is $C(x) = x^3 - 6x^2 + 400x + 1000$ (dollars, for x units). Find the marginal cost (the derivative) at $x = 10$.

Solution. $C'(x) = 3x^2 - 12x + 400$; $C'(10) = 300 - 120 + 400$.

Conclusion: marginal cost at $x = 10$ is \$580 per additional unit.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = 8x^3 - 3x^2 + 2x - 9$.

Question 2

Differentiate $f(x) = (x + 4)(2x^2 - 3)$.

Question 3

Differentiate $f(x) = \dfrac{x^2}{3x + 1}$.

Question 4

Differentiate $f(x) = (3x^2 - 1)^5$.

Question 5 — Challenge

Differentiate $f(x) = x^2(4x - 1)^3$, factoring your answer.

Question 6 — Challenge

Find the equation of the tangent line to $f(x) = x^3 - 3x$ at $x = 2$.

Question 7

Differentiate $f(x) = \sqrt{4x^2 + 9}$ (write as a power first).

Question 8 — Challenge

Differentiate $f(x) = \dfrac{(x-2)^2}{x+1}$, factoring your answer.

Question 9

Find the x-value(s) where the tangent to $f(x) = x^3 - 27x + 4$ is horizontal.

Question 10

Differentiate $f(x) = x(x-3)(x+2)$ by expanding first.

Question 11

A company's cost is $C(x) = x^3 - 9x^2 + 600x + 2000$; find the marginal cost at $x = 6$.

Question 12 — Challenge

Differentiate $f(x) = (2x+3)^2(x-1)^3$, leaving your answer factored (identify the common factors, no need to fully expand).

Question 13 — Challenge

Find the equation of the tangent line to $f(x) = \dfrac{x}{x+2}$ at $x = 0$.

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = 8x^3 - 3x^2 + 2x - 9$.
2. Differentiate $f(x) = (x+4)(2x^2-3)$.
3. Differentiate $f(x) = \dfrac{x^2}{3x+1}$.
4. Differentiate $f(x) = (3x^2-1)^5$.
5. Differentiate $f(x) = x^2(4x-1)^3$, factoring your answer.
6. Find the equation of the tangent line to $f(x) = x^3 - 3x$ at $x = 2$.
7. Differentiate $f(x) = \sqrt{4x^2+9}$ (write as a power first).
8. Differentiate $f(x) = \dfrac{(x-2)^2}{x+1}$, factoring your answer.
9. Find the x-value(s) where the tangent to $f(x) = x^3 - 27x + 4$ is horizontal.
10. Differentiate $f(x) = x(x-3)(x+2)$ by expanding first.
11. A company's cost is $C(x) = x^3 - 9x^2 + 600x + 2000$; find the marginal cost at $x = 6$.
12. Differentiate $f(x) = (2x+3)^2(x-1)^3$, leaving your answer factored (identify the common factors, no need to fully expand).
13. Find the equation of the tangent line to $f(x) = \dfrac{x}{x+2}$ at $x = 0$.

✓ Answer Key

1. $f'(x) = 24x^2 - 6x + 2$
2. $2(2x^2-3) + (x+4)(4x) = 6x^2 + 16x - 6$
3. $\dfrac{2x(3x+1) - x^2(3)}{(3x+1)^2} = \dfrac{3x^2+2x}{(3x+1)^2}$
4. $30x(3x^2-1)^4$
5. $2x(4x-1)^3 + x^2 \cdot 3(4x-1)^2 \cdot 4 = x(4x-1)^2[2(4x-1) + 12x] = x(4x-1)^2(20x-2)$
6. $f'(x) = 3x^2 - 3$, $f'(2) = 9$; $f(2) = 8 - 6 = 2$; $y - 2 = 9(x-2) \Rightarrow y = 9x - 16$
7. $f(x) = (4x^2+9)^{1/2}$; $f'(x) = \dfrac{8x}{2\sqrt{4x^2+9}} = \dfrac{4x}{\sqrt{4x^2+9}}$
8. $\dfrac{2(x-2)(x+1) - (x-2)^2}{(x+1)^2}$; factor $(x-2)$: $(x-2)[2(x+1) - (x-2)] = (x-2)(x+4)$: $f'(x) = \dfrac{(x-2)(x+4)}{(x+1)^2}$
9. $f'(x) = 3x^2 - 27 = 0 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$
10. expand $(x-3)(x+2) = x^2 - x - 6$, so $f(x) = x^3 - x^2 - 6x$: $f'(x) = 3x^2 - 2x - 6$
11. $C'(x) = 3x^2 - 18x + 600$; $C'(6) = 108 - 108 + 600 =$ \$600
12. $u = (2x+3)^2, v = (x-1)^3$; $u' = 4(2x+3)$, $v' = 3(x-1)^2$; $f'(x) = 4(2x+3)(x-1)^3 + 3(x-1)^2(2x+3)^2$, factor $(2x+3)(x-1)^2$: $= (2x+3)(x-1)^2[4(x-1) + 3(2x+3)] = (2x+3)(x-1)^2(10x+5)$
13. $f'(x) = \dfrac{(x+2) - x}{(x+2)^2} = \dfrac{2}{(x+2)^2}$; $f'(0) = \dfrac{2}{4} = 0.5$; $f(0) = 0$; tangent: $y = 0.5x$